RLP # 36C248-25-R-0110 – EXHIBIT B

Agency's Special Requirements – Ceiba CBOC

The U.S. Department of Veterans Affairs (VA) desires to lease space for the Ceiba CBOC (replacement clinic) The space will be used as Community Based Outpatient Clinic (CBOC).

The space requested is Net Usable Square Feet (NUSF) = 12,757 which in ANSI/BOMA Office Area (ABOA) = 15,308

The space will be used as Community Based Outpatient Clinic (CBOC). On-site parking for approximately

- Eighty-Seven (87) vehicles, with a minimum of
- Eight (8) dedicated to handicapped parking spaces is required.

The Space Program is comprised of the following rooms:

General Function	Room	Net Area	NUSF	ABOA	Floor Covering	Base	Wall Covering	Ceiling
HBPC	Team Work Area, HBPC	400	600	720	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
HBPC	Utility Room, Clean	80	120	144	Welded seam sheet floor	Welded seam sheet floor	High Build Glazed Coating	Acoustical Tile
Lab	Blood Specimen Collection Room	160	240	288	Welded seam sheet floor	Welded seam sheet floor	High Build Glazed Coating	Acoustical Tile
Lab	Laboratory, General	150	225	270	Welded seam sheet floor	Welded seam sheet floor	High Build Glazed Coating	Acoustical Tile
Lab	Handwashing Station, Clncl Sprt	10	15	18	Welded seam sheet floor	Welded seam sheet floor	High Build Glazed Coating	Acoustical Tile
Lab	Toilet, Specimen Collection	60	90	108	Porcelain tile	Porcelain tile	Porcelain Tile	Acoustical Tile
Mental Health	BIHIP Provider 1 Consult	120	180	216	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Mental Health	BIHIP Provider 2 Consult	120	180	216	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Mental Health	BIHIP Provider 3 Consult	120	180	216	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Mental Health	BIHIP Provider 4 Consult	120	180	216	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Mental Health	Group Therapy Room	320	480	576	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Primary Care	Consult Room, (PACT) Dietitian	125	187.5	225	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Primary Care	Consult Room 1	125	187.5	225	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Primary Care	Consult Room 2	125	187.5	225	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Primary Care	Consult Room 3	125	187.5	225	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Primary Care	Consult Room 4	125	187.5	225	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile

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Primary Care	Exam Room 1	125	187.5	225	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Primary Care	Exam Room 2	125	187.5	225	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Primary Care	Exam Room 3	125	187.5	225	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Primary Care	Exam Room 4	125	187.5	225	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Primary Care	Exam Room 5	125	187.5	225	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Primary Care	Exam Room 6	125	187.5	225	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Primary Care	Exam Room 7	125	187.5	225	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Primary Care	Exam Room 8	125	187.5	225	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Primary Care	Exam Room 9	125	187.5	225	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Primary Care	Toilet Procedure Room	75	112.5	135	Porcelain tile	Porcelain tile	Porcelain Tile	Acoustical Tile
Primary Care	Procedure Room & Womens	180	270	324	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Bariatric Patient Toilet, Bldg Sprt	75	112.5	135	Porcelain tile	Porcelain tile	Porcelain Tile	Acoustical Tile
Support	Tele-Retinal Room	120	180	216	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Tele-Health Room	125	187.5	225	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Canteen, PACT 2	110	165	198	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Office, HAS Supervisor	100	150	180	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Office, Nurse Manager	100	150	180	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Office, CMO	100	150	180	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Toilet, Female	180	270	324	Porcelain tile	Porcelain tile	Porcelain Tile	Acoustical Tile
Support	Toilet, Male	180	270	324	Porcelain tile	Porcelain tile	Porcelain Tile	Acoustical Tile
Support	Reception	300	450	540	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Toilet, Family / Bariatric	80	120	144	Porcelain tile	Porcelain tile	Porcelain Tile	Acoustical Tile
Support	Alcove, Wheelchair/ Stretchers	90	135	162	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Vestibule	120	180	216	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Waiting, PACT 1	415	622.5	747	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Workstation, Patient Education	30	45	54	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Medication Room, Clncl Sprt	80	120	144	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Locker, Staff Personal Property	60	90	108	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile

Support	Storage, Medical Equipment	120	180	216	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Utility Room, Clean	120	180	216	Welded seam sheet floor	Welded seam sheet floor	High Build Glazed Coating	Acoustical Tile
Support	Toilet, Staff	120	180	216	Porcelain tile	Porcelain tile	Porcelain Tile	Acoustical Tile
Support	Alcove, Height / Weight Station	30	45	54	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Conference Room	320	480	576	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Training Room / Lounge	240	360	432	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Ext. Team-Caregiver	50	75	90	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Ext. Team- Pharm D	50	75	90	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Ext. Team- Suicide Prevention	50	75	90	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Ext. Team- Whole Health	50	75	90	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Ext. Team-Beh. Health Consultant	50	75	90	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Ext. Team-Homeless (HUD-V)	50	75	90	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Ext. Team-SW PCMHI	50	75	90	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Alcove, Height / Weight	40	60	72	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Alcove, Resuscitation Cart	20	30	36	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Copy / Supply Alcove,	40	60	72	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Team Work Area 1	215	322.5	387	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Team Work Area 2	215	322.5	387	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Team Work Area 3	215	322.5	387	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Operations Room	100	150	180	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Holding Room	45	67.5	81	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Storage, Secure	25	37.5	45	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Storage, Equipment	100	150	180	Luxury Vinyl Tile	Resilient Base	Paint	Acoustical Tile
Support	Communications Room	120	180	216	Antistatic Coating or material	Backboard	Backboard	Exposed
Support	Utility Room, Soiled	80	120	144	Welded seam sheet floor	Welded seam sheet floor	High Build Glazed Coating	Acoustical Tile

Support	Housekeeping Aides Closet	60	90	108	Quarry tile	Quarry tile	High Build Glazed Coating	Acoustical Tile
		8,505	12,757	15,308				

The administrative/common spaces are described as follows:

RECEPTION AREA

- Data outlets by workstation, One (1) quadraplex, location TBD by desk configuration
- Electrical outlets by workstation, One (1) quadraplex, location TBD by desk configuration
- Electrical outlets, remaining walls, One (1) every 6 (linear) Feet, as established by Building Codes.
- Duress Alarm, hard wired. Reports to stand-alone system.
- Luxury Vinyl Tile (LVT).
- Direct access to waiting area.
- Window Coverings to have horizontal manual roller shades with solar glare and heat reduction like Mecho Shade, Alkenz or similar product.
- Centralized check-in/check-out for efficient utilization of staff.
- Solid Surface Countertop, minimum eight (8) foot, that utilizing concealed mounting brackets where practical

OFFICES / CONSULT ROOMS / EXAM ROOMS

- Data outlets by workstation, One (1) quadraplex, location TBD by desk configuration
- Electrical outlets by workstation, One (1) quadraplex, location TBD by desk configuration
- Electrical outlets, remaining walls, One (1) every 6 (linear) Feet, as established by Building Codes.
- Window Coverings to have horizontal manual roller shades with solar glare and heat reduction like Mecho Shade, Alkenz or similar product. **
- Sound Transmission Class (STC) rating of 50. The sound resistant enclosures shall be designed to assure speech privacy and achieve an STC50. *(Examples: sound mitigating ceiling tiles, walls 12-18" above ceiling, ROXUL AFB Sound Attenuation Ceiling Insulation 3", etc.).*
- Solid Core Wood Doors to have sound gaskets and door sweeps for sound attenuation.
- One (1) Coat hook behind the door
- Positioning furniture to ensure safety the of clinical staff; provided easy egress or exit. Reference the conceptual drawing for furniture placement.
- Consult rooms and exam rooms to accommodate wheelchair and furniture arrangement
- Consult room lay-out need to provide adequate furniture lay-out so provider has quick and clear path to the door.
- All rooms (except Telecommunications Room) should be master keyed, locking mechanism should be on the inside of the frame, and locking mechanism must be approved by the VA.

WAITING AREA

- Electrical outlets, One (1) every 6 (linear) Feet
- Smoke Detector. Hardwire Combination Smoke and Carbon Monoxide Smoke Alarm with Battery Backup and Voice Alert.
- Luxury Vinyl Tile (LVT).
- Luxury Vinyl Tile (LVT) to continue through hallway/corridors.
- Direct window access to waiting area.
- Window Coverings to have horizontal manual roller shades with solar glare and heat reduction like Mecho
- Shade, Alkenz or similar product.
- Access Controlled Door to counseling clinical area. Wood or Steel door with a full-length glass inserted panel.
- Corridors shall be at minimum of 6 feet in clear width.

- Chair Rail (28 to 32 inches) on walls in waiting area
- Vinyl Corner Guards (from top of Vinyl Base) mounted at height 48” in hallways.
- Drinking Fountain. A wall hung stainless Bottle Filling Station & Single ADA Cooler provided in the waiting area. Maintenance and filter replacement by the Lessor.
- Cable TV audio/video wall jack, HDMI, dual Data outlet and quadruplex Electrical outlet. Wall backing and mounting materials for connection to a VA furnished television 50” to 60”; Lessor installed. Mounting heights for power and data to be coordinated with the Vet Center Director during design.
- Cable Television Service paid for/provided by the Lessor through the annual rent in 4 locations. (Waiting, Group Rooms and Family Multi-Purpose Room).
- Guest high speed internet Wi-Fi at or above 600 Mbps paid for/provided by the Lessor through the annual rent

EMPLOYEE BREAK ROOM
- Six (6) duplex Electrical outlets (separate circuit for refrigerator and for microwave oven on , two counter height (GFFI), two 18” from floor).
- Luxury Vinyl Tile (LVT).
- Perimeter Walls shall extend from floor slab to six inches above the ceiling grid and filled with fiberglass bat insulation to ensure acoustical privacy.
- Smoke Detector - Hardwire Combination Smoke and Carbon Monoxide Smoke Alarm with Battery Backup and Voice Alert.
- Kitchen Cabinets –upper and lower kitchen cabinets of high-quality plastic laminate or stained wood (particle board or MDF is not acceptable)
- Kitchen Countertop - minimum ten (10) feet a solid surface countertop like Corian.
- Sink - double compartment stainless sink.
- Garbage Disposal.
- Dishwasher provided by the Lessor.
- Built-in Convection Microwave Oven provided by the Lessor.
- Refrigerator (20-25 cu. ft.) provided by the Lessor.
- Water feed for Ice Maker on refrigerator.
- Solid Core Wood Doors to have sound gaskets and door sweeps for sound attenuation.
- Door shall have an automatic closure installed and deadbolt lock according to VA security regulations

JANITORIAL / HOUSEKEEPING AID CLOSET (HAC)
- Two (2) GFCI duplex Electrical outlets.
- Sealed concrete floor.
- Adjustable wire shelving on two walls.
- Mop Sink.

ADDITIONAL BUILDING/DESIGN REQUIREMENTS:

Preventive maintenance of building systems such as HVAC, domestic water heaters, and electric panel boards will be completed prior to occupying the space and throughout the occupancy of the building to meet Joint Commission Accrediting standards.

Special care should be taken to avoid storing construction materials in non-climate-controlled environments.

Clean entire space (aka “terminal cleaning”) prior to VA occupying space.

Building’s Interior Construction Material i.e. finishes shall be sent to the VA interior designer and the VA Activations Teams for color selection review and concurrence. Furniture placement should be discussed reviewed with them as well. In addition, these teams are to review location of location and color selection. Chair Rail or similar wall protection (Acrovyn strip) and cubicle (hospital) curtain tracks layouts

Natural (sun) light is highly desirable, specially in main waiting area and entrance.

HEATING VENTILATION AND COOLING HVAC

- Common (overall) building temperature, between 75 to 70 degrees Fahrenheit, with relative humidity between 60% and 30%.
- SEE APPENDIX A for rooms with specific requirements, validate information VA's HVAC Design Manual

Site/Civil
Pedestrian and Vehicle Access & Circulation

- Minimum traffic lane width is 12 feet, and minimum sidewalk width is 4 feet. Curves for traffic lanes and radii at intersections must be adequately sized to prevent vehicles from encroaching on an opposing lane of traffic.
- An accessible route must be provided from the public right-of-way abutting the site to the accessible building entrance.
- Provide a service area with a loading dock designed to accommodate truck (WB-62) maneuverability. Loading dock shall be 4 feet above the driveway. Platforms shall have a minimum depth of 8 feet front to back or between dock lift/leveler and back wall. Provide canopy over the platform with 14 feet of clearance from grade to the underside of the canopy. Canopy shall extend minimum of 4 feet beyond the edge of the dock for weather protection. Provide stair or ramp to the platform. Provide hydraulic dock levelers with 25,000 pounds capacity for recessed installation at loading dock.
- Service area shall accommodate vehicles that pick-up trash and recycled materials. Locate service area away from public and patient areas.
- Provide reflective traffic control signs as required for intersections, no parking lanes, and guidance of site traffic.
- Where applicable (if site allows):
    - provide concrete and (ornamental) wrought iron Exterior fencing similar to the one that VA San Juan, Main Campus
    - provide entrance signage wall (for building to be recognized from street) like the one at the VA San Juan.

Additional Site/Civil Requirements:

- A pickup/drop off area located at the Main 'Public' Entrance. The pick-up/drop-off shall have canopy cover designed to accommodate public bus, ambulances, and shuttle services. The canopy shall be a noncombustible permanent structure. Final determination of canopy height and location shall be coordinated between the A/E and medical center.
- All concrete curbs must be painted high visible yellow and maintained as such. At minimum, all curbs shall be repainted every three years, or as needed.
- Provide necessary curb cuts with truncated domes at all major pedestrian intersections with vehicle pathways.

Entry Canopies

- Provide non-combustible canopies over the following locations: patient entry to clinic, ambulance building access point and receiving area/loading dock.
- At patient entry, provide a covered patient drop-off zone with space for at least one full

size passenger vehicle (19 feet long) and an accessible access aisle.

- At ambulance building access point, provide a covered patient drop-off zone with space for at least one ambulance (typical size used in area) and an accessible access aisle.
- Provide clearance from grade to underside of canopy with for typical emergency and transport vehicles used in the area.
- Final determination of canopy height and location shall be coordinated between the A/E and medical center.

Parking

- Parking lots with 90-degree stalls must have minimum parking stall dimensions of 9 feet by 18 feet, and a minimum drive aisle width of 24 feet. Angled parking must have one-way drive aisles with the same stall sizes as 90-degree parking. Angled parking drive aisle width must comply with a published design standard for a designated parking angle.

Additional Parking Requirements:

Parking spaces should accommodate 87 cars. This includes 8 handicap spaces.

Parking, both handicapped and regular must be readily available and marked VA ONLY.

Site Grading

- Roads and walks should have a typical cross slope of 2% unless adequate surface drainage is provided by other slope conditions.

Standby Generator

(requirements in the absence of a code required Essential Electrical System)

- Packaged Engine Generator
  - An exterior diesel fuel generator, in weatherproof sound attenuated enclosure, kW/kVA size as determined for loads noted below, will be provided supplying emergency power for the facility.
  - Engine: NFPA 37 compliant.
  - Cooling System: Closed-loop, liquid-cooled, radiator mounted on generator set base.
  - Fuel Tanks: 24 hour run time at full load sub-base tank
  - Engine Exhaust System: Critical silencing muffler.
  - Combustion Air-Intake System: Filter type air intake silencer, intake duct and connections.
  - Starting System: Electric with negative ground.
  - Automatic Transfer Switches: 4-pole switches are required.

Additional Requirements:

- The installed generator shall be capable of providing 100% full power (normal and emergency) with on-site fuel storage capacity to provide power for at least 96 hours (4 days).
- Lessor shall have in place a contract or MOA for priority fuel delivery to the clinic. Monthly Load Test shall be performed by Lessor and shall keep written records/log

Millwork

Refer to Room Finishes, Door, & Hardware Schedule PG 18-14, and requirements as follow:

Use solid surface countertops in wet areas and laminates in areas that do not have risk of water damage.

Solid Surface Fabrications: Hard, solid, non-porous mineral-filled acrylic resin (methyl methacrylate) material in color to be selected by designer. Thickness: 1/2” minimum.

- Locations:
- Transaction & Counters tops at nurse stations and reception desks.
- All counters in patient treatment areas and breakroom/lounge to have solid surface countertop with 4” high minimum integral coved side and backsplash.
- Where scheduled as wall protection, provide ¼” thick material with welded seams.

Additional Equipment Requirements:
MDF and particle not acceptable for millwork.

Acoustics & Speech Privacy

Refer To Pg 18-3 – Design And Construction Procedures, Topic Number 11, Noise Transmission Control

Equipment

Refer to PG-18-5 Equipment Guide List, and specifically to UTUADO’s Initial Room Content List, provided with the RLP for equipment, accessories, and furnishings descriptions and locations. Items to be provided and/or installed by Lessor are noted on the list. Provide partitions, partition backing, and above ceiling structural support as required for wall and/or ceiling mounted equipment, accessories, and furnishings. Provide required utility connections for scheduled equipment.

Additional Equipment Requirements:
Lessor is responsible for providing all utilities and rough-ins to government furnished equipment as well as coordinating these with the Activations team.

Locks/Hardware:

Refer to Room Finishes, Door, & Hardware Schedule PG 18-14, and requirements as follow: Provide extra heavy duty, Grade 1 hardware for all components.

Major components and finishes as listed below:

- Cylinders and Keying: Key locks/cylinders in groups with new master key or grandmaster key system as directed by Owner. Provide three (3) keys per lock. Provide construction master keying. Cylinders shall meet the requirements of ANSI/BHMA A156.5-14.
- Low Energy Automatic operators, ANSI A156.19-07. Heavy duty commercial grade. Provide complete with drop plates, bracket, or adapters for arms as required to suit details. Provide a terminal strip in an enclosed box near or above door that indicates connections for Security and Fire Alarm equipment and for electrified hardware items associated with proper door operation, as indicated by hardware group operational description. Refer to floor plans for type of actuation devices and bollards if required. Coordinate with Security Contractor for doors actuated by electronic access control system.
- Electronic access control: Electronic access control system/device(s), power supplies (unless otherwise noted in hardware group) and monitoring/alarm(s) are provided with Security System. General Contractor to coordinate the provision and installation of the products. Refer to documents with Security Information for location(s) and type(s) of control(s). Connection by Electrical

- *Additional Requirements:*

Stanley/best 7 Pin Small format cores. A new Great Grandmaster key shall be established for this project. The key system shall be removable core type. The key blanks shall be protected by a utility patent with a minimum seven years remaining on the patent from the start of construction and protected by contract-controlled distribution. Provide two keys for each lock. The name of the manufacturer, or trademark by which manufacturer can readily be identified, shall be legibly marked on each lock, the key change number shall be marked on the exposed face of lock, and stamped on each key. Key change numbers shall provide sufficient information for replacement of the key by the manufacturer. The manufacturer shall furnish code pattern listings in both paper and electronic formats so keys may be reproduced by code.; provide electronic format in file type required by project's key control software. The manufacturer shall design the new key system with the capacity to rekey the system and provide for 25 percent expansion capability beyond this requirement. Submit a keying chart for approval showing proposed keying layout and listing expansion capacity. Keying information will be furnished to the Contractor by the COR. Supply information regarding key control of cylinder locks to manufacturers of equipment having cylinder type locks. Notify COR immediately when and to whom keys or keying information is supplied. Return all such keys to the COR

Telecommunications

- The Lessor shall provide the following:
  - Telephone cabling, pathways (conduit and cable tray), outlets, faceplates, terminal blocks, backboards, cable terminations and cable testing.
  - Data cabling (fiber optic and copper), pathways (conduit and cable tray), outlets, faceplates, patch panels, network equipment racks, network equipment cabinets, cable terminations and cable testing.
- The VA IT department will provide the following:
  - Telephone System hardware and electronics such as voice mail servers and telephone handsets.
  - Data network electronics such as concentrators, Ethernet switches, servers, PCs, Wireless Access Points and other electronic equipment.

- Pathways
  - Boxes and Conduits
    - Voice and data outlets shall be provided with a 4" square by 2-1/8" deep box with a single gang, telecommunication rated work box with a minimum of a ¾" conduit routed up to an accessible ceiling space.
  - Sleeves
    - Where cables penetrate through walls, conduit sleeves with bushings on both ends, shall be provided. All penetrations through fire rated walls shall be fire stopped.
    - Where cables penetrate through floors of telecommunications rooms, a minimum of four (4) 4-inch conduit sleeves with bushings on both ends, shall be provided. All penetrations through floors shall be fire stopped.
    - Conduit sleeves shall be sized to be filled with cables to no more than 40 percent of the cross-sectional area of the conduit.
  - Cable Support
    - Wire mesh cable tray a minimum of 4" deep x 12" wide shall be provided and supported from the structural steel or concrete structure with a minimum of 3/8" diameter threaded rods to support the horizontal and backbone communications cables along the main pathways above the suspended ceiling space. In finished spaces without a suspended ceiling, provide a minimum of 4" deep x 12" wide Solid Bottom cable tray instead of Wire Mesh cable tray along the main pathways. Cable trays shall be sized to be filled with cables to no more than

50% of the cross-sectional area of the cable tray. Where the cable tray fill ratio exceeds 50% of the cable tray cross-sectional area, provide a larger cable tray or two cable trays.
        - A minimum of 12 inches of free access shall be provided and maintained above the cable trays and along one side of the cable trays.
        - Where cables are routed in the open outside of the cable tray above the suspended ceiling space, adequate cable support via J-hooks shall be located at a maximum of 48" intervals.
        - As per National Electrical Code and TIA-569 standard, the suspended ceiling support wires or support rods shall not be used as a means of cable support. Cables shall not be laid directly on the ceiling tile, ceiling grid rails, or on the structural steel (bar joists). An independent hanger system shall be used.

- Spaces
    - Entrance Facility Room (AKA Demarcation Point- Demarc)
        - The Entrance Facility Room is the location where the Local Exchange Carriers and other communications Service Providers such as telephone, data, and MATV/CATV install their cabling and equipment to bring services into the building. It also establishes the physical point where the service provider's responsibilities for service and maintenance end.
        - The minimum size of the Entrance Facility Room shall be 80 sq. ft. with an additional 20 sq. ft. for every additional rack required.
        - The room shall not be located directly below or adjacent to laboratories, kitchens, laundries, rest rooms, showers, or other facilities where water service is provided.
        - Any pipe or duct system foreign to the room installation shall not enter or pass through the room. The design professional shall ensure that foreign piping such as water pipes, steam pipes, medical gas pipes, sanitary waste pipes, roof drains, AC ducts, and other unrelated piping containing liquids or gases are not installed or pass through the room. Sprinkler piping shall not be routed through the room, unless it serves to protect the installation.
        - A minimum of (2) 4" conduits shall be installed out to the property line to provide pathways for the services providers to install their cabling.
        - All the walls of the room shall be constructed from drywall deck to deck, not just from floor to suspended ceiling height. All the walls of the room shall be covered from the floor to a minimum height of 8'-0" above the floor with 3/4-inch exterior AC grade flame retardant plywood and painted a light color to reflect the room light and reduce dust.
        - The lighting shall be a minimum of 500 lux in the horizontal plane and 200 lux in the vertical plane when measured at 3 feet above the finished floor.
        - The door shall be a minimum of 36 inches wide and 96 inches high, hinged to open outward and fitted with a card reader security lock.
        - Protective cages shall be installed on all water-based fire protection sprinkler heads located within the room.
        - The Service Provider IT equipment installed in the Entrance Facility Room will be required to operate 24 hours a day and 365 days a year. The HVAC system shall be designed and installed to maintain a room temperature of 64-75 degrees Fahrenheit and relative humidity of 30-55 percent noncondensing on a 24-hour basis.
- Main Computer Room (MCR)
    - The Main Computer Room is a centralized space for telecommunications and

computer equipment that serves an entire building. Typical equipment includes phone switches, voice mail servers, file/application servers, video surveillance storage and Core Ethernet switches. The requirement for an MCR will be dependent upon the size and function of the CBOC.

- The location of the Main Computer Room should be determined after careful consideration. Locations should be avoided that restrict expansion of the room due to building construction such as elevators, mechanical rooms, core hallways, outside walls, or other fixed building walls. The location should consider accessibility requirements for the delivery of large equipment to the room and be located away from EMI sources that limit EMI field strength to no more than 3.0 V/m throughout the frequency spectrum.
- The room shall not be located directly below or adjacent to laboratories, kitchens, laundries, rest rooms, showers, or other facilities where water service is provided.
- Any pipe or duct system foreign to the room installation shall not enter or pass through the room. The design professional shall ensure that foreign piping such as water pipes, steam pipes, medical gas pipes, sanitary waste pipes, roof drains, AC ducts, and other unrelated piping containing liquids or gases are not installed or pass through the room. Sprinkler piping shall not be routed through the room, unless it serves to protect the Main Computer room installation.
- The Main Computer Room shall be dedicated to telecommunications and computer equipment. The room shall not be shared with electrical equipment, heating/ventilating and air conditioning equipment, fire detection systems, or other mechanical systems unless these systems are specifically needed and dedicated to support the computer room and its functions.
- The minimum sizes of the Main Computer Room shall be as follows:
  - For VA-occupied space less than 25,000 sq. ft. the Telecommunications Enclosure (TE) or Telecommunications Room (TR) will serve as the Main Computer Room.
  - For VA-occupied space between 25,001 sq. ft. and 3000,000 sq. ft.), no Main Computer Room shall be provided. Equipment will be placed in Telecommunications Rooms (TRs). This main Telecommunications Room (TR) shall be sized as follows:
    - 25,001 sq ft. – 50,001 sq. ft. shall be 170 sq. ft.
    - 50,001 sq. ft. – 100,000 sq. ft. shall be 190 sq. ft.
    - 100,001 sq. ft. - 150,000 sq. ft. shall be 210 sq. ft.
    - 150,001 sq. ft. – 200,000 sq. ft. shall be 230 sq. ft.
    - 200,001 sq. ft. - 250,000 sq. ft. shall be 250 sq. ft.
    - 250,001 sq. ft. – 300,000 sq. ft. shall be 270 sq. ft
  - For VA-occupied space greater than 300,000 sq. ft. or clinics that include surgery, the Main Computer Room shall be 780 sq. ft. Follow the *Generic Extra Small Campus Support Center (Data Center) Design Portfolio*. The clear height for this Main Computer Room must be a minimum of 12 feet (16 recommended).
- All the walls of the Main Computer Room shall be constructed of drywall deck to deck, not just from the floor to the suspended ceiling height. The floor, walls, and ceiling shall be sealed/painted to reduce dust and shall be light colored to reflect room light.
- Flooring materials shall be used that have antistatic properties.
- At least one of the walls of the room shall be covered from the floor to a minimum

height of 8’-0” above the floor with 3/4-inch exterior AC grade flame retardant plywood and painted a light color to reflect the room light and reduce dust.
    - 24” wide ladder racking cable tray shall be provided and supported from the structure above with a minimum of 3/8” diameter threaded rods over the rows of equipment racks/cabinets to support the horizontal and backbone communications cables through-out the Main Computer Room.
    - The lighting shall be a minimum of 500 lux in the horizontal plane and 200 lux in the vertical plane when measured at 3 feet above the finished floor in between all rows of equipment cabinets and equipment racks.
    - The door shall be 36 inches wide and 96 inches high and fitted with a card reader security lock. Where the *Generic Extra Small Campus Support Center (Data Center) Design Portfolio* has been specified, this shall be a double door 72 inches wide and 96 inches high, without a central mullion.
    - Protective cages shall be installed on all water-based fire protection sprinkler heads located within the Main Computer room.
    - The IT equipment installed in the Main Computer Room will be required to operate 24 hours a day and 365 days a year. The HVAC system shall be designed and installed to maintain a room temperature of 72-81 degrees Fahrenheit and relative humidity of 6-60 percent non-condensing on a 24-hour basis.
        - Where a TE or TR is used as the Main Computer Room for the facility, the environmental condition requirements are 64-75 degrees Fahrenheit and relative humidity of 30-55 percent.
- Telecommunications Room (TR)
    - There shall be at least one Telecommunications Room on each floor. Each work area shall be served by a Telecommunications Room that is located on the same floor that the work area is located. There shall be a minimum of one (1) Telecommunications Room in each building.
    - The Telecommunications Room shall not be located directly below or adjacent to laboratories, kitchens, laundries, rest rooms, showers, or other facilities where water service is provided.
    - Any pipe or duct system foreign to the Telecommunications Room installation shall not enter or pass through the room. The design professional shall ensure that foreign piping such as water pipes, steam pipes, medical gas pipes, sanitary waste pipes, roof drains, AC ducts, and other unrelated piping containing liquids or gases are not installed or pass through the room. Sprinkler piping shall not be routed through the Telecommunications Rooms, unless it serves to protect the installation.
    - The location of the Telecommunications Rooms shall be as close as possible to the central core of the building floor to keep horizontal cable lengths to a minimum. Additional Telecommunications Rooms shall be provided where the horizontal cable length from the telecommunications room to the farthest workstation location exceeds 90 meters (295 feet).
    - Telecommunications Rooms located on the same floor shall be no farther than 150 meters (500 feet) apart to limit horizontal cable lengths to 90 meters (295 feet) or less.
    - The Telecommunications Room shall be dedicated to telecommunications facilities and function. The room shall not be shared with electrical equipment, heating/ventilating and air conditioning equipment, or other mechanical systems unless these systems are specifically needed and dedicated to support the Telecommunications Room and its functions.
    - The Telecommunications Rooms on each floor shall be vertically aligned between

floors in a multistory building to allow for the ease of installing vertical backbone cabling.

  - The minimum sizes of the Telecommunications Rooms shall be as follows:
    - For facilities from 1 to 3,000 sq. ft., provide a standard 26U Telecommunications Enclosure located in a secured space that is not necessarily dedicated to telecommunications purposes. See sheets 16-19 of the *CBOC Inside Plant Information Transport Systems Specification*.
    - For facilities from 3,001 to 6,000 sq. ft., provide a 1-rack TR (80sf, 10'x8' form factor). See sheet 6 of the *CBOC Inside Plant Information Transport Systems Specification*.
    - For facilities from 6,001 to 10,000 sq. ft., provide a 2-rack TR (100sf, 10'x10' form factor). See sheet 6 of the *CBOC Inside Plant Information Transport Systems Specification*.
    - For facilities from 10,001 to 25,000 sq. ft., provide a 3-rack TR (120sf, 10'x12' form factor). See sheet 6 of the *CBOC Inside Plant Information Transport Systems Specification*.
    - For facilities greater than 25,000 sq. ft., provide TRs in addition to the MCR per the sizing and quantity determined by the *TR Design Checklist* in accordance with the serving zone size of each planned TR.
  - The above sizes should be confirmed with specific IT equipment sizes. The above sizes also assume there are no obstructions in the room such as columns and the rooms are a rectangular shape.
  - All of the walls of the Telecommunications Rooms shall be constructed from drywall deck to deck, not just from floor to suspended ceiling height. A minimum of 3 of the walls of the Telecommunications Room shall be covered from the floor to a minimum height of 8'-0" above the floor with 3/4-inch exterior AC grade flame retardant plywood and painted a high gloss white with two coats of fire-resistant paint. Reserve a minimum of 12" dedicated space in front of the walls to accommodate equipment being mounted to the wall.
  - The floor shall be covered with light colored luxury vinyl floor tile to reflect the room light and reduce dust.
  - 18" wide ladder racking cable tray shall be provided and supported from the structure above with 3/8" diameter threaded rods over the equipment racks and the side walls to support the horizontal and backbone communications cables through-out the Telecommunications Room.
  - The lighting shall be a minimum of 500 lux in the horizontal plane and 200 lux in the vertical plane when measured at 3 feet above the finished floor.
  - The door shall be a minimum of 36 inches wide and 96 inches high, hinged to open outward and fitted with a card reader security lock.
  - Protective cages shall be installed on all water-based fire protection sprinkler heads located within the telecommunications rooms.
  - The IT equipment installed in the Telecommunications Room will be required to operate 24 hours a day and 365 days a year. The HVAC system shall be designed and installed to maintain a room temperature of 64-75 degrees Fahrenheit and relative humidity of 30-55 percent non-condensing on a 24-hour basis.

- Telecommunications Bonding and Grounding
  - Telecommunications Primary Bonding Busbar
    - Each Entrance Facility Room shall contain a Telecommunications Primary Bonding busbar for providing a central location for bonding all telecommunications equipment

in the Entrance Facility Room per the TIA-607-D Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises, local codes, the VA Electrical Design Manual, and National Electrical Safety Code.

  - The Telecommunications Primary Bonding busbar shall consist of a predrilled copper busbar with TIA-607 standard sizing and spacing. It shall have minimum dimensions of ¼ inch thick, 4 inches wide, and the length shall be a minimum of 23 inches. The bonding busbar shall be insulated from its support by a minimum of a 2-inch separation.
  - Building structural steel (beams and/or columns) within 6 feet of the bonding busbar shall be bonded to the bonding busbar with a minimum of a 6 AWG copper conductor.
- Telecommunications Secondary Bonding Busbar
  - Each Main Computer Room and Telecommunications Room shall contain a Telecommunications Secondary Bonding Busbar for providing a central location for bonding all telecommunications equipment in the room per the TIA-607 standard.
  - The Telecommunications Secondary Bonding Busbar shall consist of a predrilled copper busbar with TIA-607 standard sizing and spacing. It shall have minimum dimensions of ¼ inch thick, 2 inches wide, and the length shall be a minimum of 12 inches. The bonding busbar shall be insulated from its support by a minimum of a 2-inch separation.
  - Building structural steel (beams and/or columns) within 6 feet of the bonding busbar shall be bonded to the bonding busbar with a minimum of a 6 AWG copper conductor.
- Telecommunications Rack Bonding Busbar
  - Racks located in the Entrance Facility Room, Main Computer Room and the Telecommunications Rooms, as well as Telecommunications Enclosures (TEs) shall have a horizontal Rack Bonding Busbar installed in the top of the rack/cabinet in RU 45 (install in the top rear position in TEs) to provide effective bonding of the rack/TE to the Primary bonding busbar or Secondary Bonding Busbar and provide a central location for the bonding of all telecommunications equipment located in the rack/TE per the TIA-607 standard. The busbar shall consist of a pre-drilled copper busbar with TIA-607 standard sizing and spacing.
  - The Rack Bonding Busbar shall be bonded to the Telecommunications Primary Bonding busbar or Telecommunications Secondary Bonding busbar in the room with a minimum of a 6 AWG copper conductor.
  - Rack mounted IT equipment with integral bonding terminals shall be bonded to the Rack Bonding Conductor (RBC) or to a vertical/horizontal Rack Bonding Busbar (RBB). An RBC is a bonding conductor from the rack or RBB to the TEBC. Each cabinet or equipment rack will have a suitable connection point to which the bonding conductor can be terminated. Properly sized listed two-hole compression lugs or listed terminal blocks with two internal hex screw or equivalent torque characteristics shall be used at the connection point.
- Telecommunications Bonding Backbone Cable
  - The Telecommunications Primary Bonding Busbar in the Entrance Facility room and Telecommunications Secondary Bonding Busbars in the Main Computer room and Telecommunications Rooms shall be bonded to the building grounding electrode system with a bonding backbone cable that is a minimum of a 3/0 AWG stranded copper conductor.
  - The building structural steel shall not be used as a replacement for the bonding backbone cable.

- Bonding of Cable Tray and Equipment
  - Cable tray and/or Ladder racking shall be bonded to the Primary Bonding Busbar or Secondary Bonding Busbar with a minimum of an insulated #6 AWG stranded copper conductor and connectors designed for the specific purpose.
  - Bonding of other telecommunications equipment in the Telecommunications Rooms or Main Computer Room to the bonding busbars shall be executed as required by the equipment manufacturer.

- Equipment Racks and Equipment Cabinets
  - Ensure that IT equipment racks are installed flush to one another without air gaps between the racks. Use appropriate materials to fill gaps between the racks to prevent recirculation of exhaust air to the cold aisle. Meet the following requirements:
    - Style: Channel
    - Height: 84 inches, Width: 24 inches, Depth: 30 inches minimum.
    - Equipment Mounting Width: 19 inches.
    - Equipment Mounting Height: 45 RUs.
    - Front and Rear rails: EIA threaded or Square Holes for cage nuts.
    - Rail Marking: Rack unit markings present on front and rear rails starting at one RU at the bottom.
    - Weight Capacity: 2,500 lbs. minimum.
    - Cable Management: Built-in overhead water fall and cable management strap attachment points.
    - Seismic bracing where required by Code.
    - Provide rack PDU brackets.
    - Color: White.
  - Equipment Cabinets shall be installed in the Main Computer Room for the housing of the Server equipment meeting the following requirements:
    - Style: Enclosed equipment cabinet with side panels and front and rear doors.
    - Height: 84 inches, Width: 24 inches, Depth: 48 inches maximum with all doors and accessories installed.
    - Equipment Mounting Width: 19 inches.
    - Equipment Mounting Height: 45 RUs.
    - Front and Rear rails: Square Holes for cage nuts. Toolless adjustable.
    - Rail Marking: Rack unit markings present on front and rear rails starting at one RU at the bottom.
    - Weight Capacity: 2,500 lbs. minimum.
    - Front Door: Single perforated, minimum of 63% open.
    - Rear Door: Single solid OR Split, perforated where vertical exhaust ducts cannot be implemented.
    - Latches: Keyed lock upgradeable to keyless system compression latch.
    - Top panel: Vertical exhaust duct (heat containment) and high-capacity cable access with brush grommets.
    - Side Panel: Solid, Locking.
    - Bottom Panel: Solid with high-capacity cable access with brush grommets or air dam foam.
    - Seismic bracing where required by Code.
    - Accessories: Zero U vertical single mount PDU brackets, castors for safe movement of cabinet, leveling legs, and air dam/sealing kit.
    - Color: White.
  - Telecommunications Enclosures (TEs) installed shall meet the following requirements:

- NEMA-12 or equivalent construction. Dust seals and replaceable inlet/outlet filters for vents/airflow openings/fans provided. This is required regardless of planned installation environment.
- Filters shall be commercially widely available and initially provided with the TE.
- Environmentally controlled enclosures are acceptable.
- 24” minimum width to allow for power and telecommunications cabling management to the sides of rack-mounted equipment.
- 24RU in height or larger.
- Unit mounts to ¾” plywood backboard via 16” OC mounting for standard stud construction.
- Unit opens in rear (swings open) for access to rear of installed equipment. Unit opens in front (swinging front door) for access to front of installed equipment. Both sections are able to be physically locked.
- Adjustable 19” EIA/TIA rack rails. Rear rail kits are required.
- Top and bottom knockouts for cable/conduit entry. All knockouts must be sealable and sealed for liquid and dust entry resistance. The use of a knockout kit to create larger penetrations is acceptable.
- 115V fans to remove heat generated in TE are required. Whether these are used as exhaust, intake, or both is not specified.
- Provide TEs with fiber distribution cabinets, fiber cassettes, UTP patch panels, horizontal cable management units, and shelves as required for the specific implementation.

- Power Distribution Units (PDUs) and Uninterruptable Power Supplies (UPSs)
  - Equipment Rack and Equipment Cabinet 120/208 Volt PDUs for TRs
    - Input: 20 Amp Three-phase; 120/208V, L21-20P Plug.
    - Circuit Breakers: 3 x 2 Pole 20 Amp Hydraulic Magnetic breakers.
    - Receptacles: (30) C13 receptacles 208 Volt, (6) C19 receptacles 208 Volt, (2) 5-20 receptacles 120 Volt.
    - IP and Serial monitoring.
    - Ethernet, USB, and Environmental sensor ports.
    - Mounting: Vertically on the rear rails of the rack.
    - Quantity: Provide two (2) PDUs in each equipment rack in Telecommunications Rooms.
  - Equipment Rack and Equipment Cabinet 120/208 Volt PDUs for MCRs
    - Input: 20 Amp Three-phase; 120/208V, L21-20P Plug.
    - Circuit Breakers: 3 x 2 Pole 20 Amp Hydraulic Magnetic breakers.
    - Receptacles: (30) C13 receptacles and (6) C19 receptacles, 208 Volt.
    - IP and Serial monitoring.
    - Ethernet, USB, and Environmental sensor ports.
    - Mounting: Vertically on the rear rails of the rack.
    - Quantity: Provide two (2) PDUs in each equipment rack and each equipment cabinet in Main Computer Room.
  - Equipment Rack and Equipment/Cabinet UPSs for TRs and MCRs
    - Input: 20 Amp Three-Phase; L21-20P Plug.
    - Output: One (1) L21-20R receptacle
    - Capacity: 5 kW
    - Run time at full capacity: Minimum of 10 minutes.
    - Mounting: Rack or Cabinet 19-inch TIA-310 mounting width.
    - Quantity: Provide one (1) UPS in each equipment rack and each equipment cabinet

in Main Computer Room and Telecommunications Rooms. (No rack-mounted UPSs are installed in the Extra Small Campus Support Center (Data Center) specified for VA-occupied spaces greater than 50,000 sq. ft.)

- Equipment Rack and Equipment/Cabinet UPSs for TEs
  - Input: 20 Amp single-phase L5-20P Plug.
  - Output: One (1) L5-20R receptacle
  - Capacity: 2 KW
  - Run time at full capacity: Minimum of 10 minutes.
  - Mounting: Rack or Cabinet 19-inch TIA-310 mounting width.
  - Quantity: Provide one (1) UPS in each Telecommunications Enclosure (TE).
- Zone Power Distribution Units (PDUs)
  - Telecommunications Rooms and Main Computer Rooms with more than one equipment rack and/or more than one equipment cabinet, shall be provided with Zone PDUs used for power distribution to the rack mounted and cabinet mounted PDUs and UPSs.
  - Input: Two (2) 30 Amp Three-Phase L21-30P plugs. Power cords on the Zone PDU shall be of sufficient length to reach the supply branch circuit receptacles suspended over the rack or cabinet.
  - Output: Four (4) L21-20R receptacles.
  - Quantity: One (1) Zone PDU for every two (2) equipment racks. One (1) Zone PDU for every two (2) equipment cabinets.
  - Supply Branch Circuits: Provide two (2) 30 amp 3-phase 120/208 Volt (Wye) circuits with L21-30R receptacles for each Zone PDU. If a Generator is installed at the site, connect the branch circuits to a Panelboard connected to the Generator. Suspend the receptacles over the equipment racks/cabinets from the ceiling for each Zone PDU.
- Equipment Rack and Equipment Cabinet 120 Volt PDUs for TEs
  - Input: 20 Amp 120 Volt, NEMA L5-20P Plug.
  - Receptacles: Minimum of eight (8) 5-15/20R receptacles.
  - Mounting: Horizontal in Rack or Cabinet 19-inch TIA-310 mounting width.
  - Quantity: Provide 2 horizontal rackmount PDUs when a Telecommunications Enclosure (TE) is specified.

- Telecommunications Infrastructure Plant (TIP)
  - Horizontal Cabling
    - Cable
      - The horizontal cabling shall consist of a minimum of two (2) Category 6A UTP LP rated cables to each work area outlet for voice and/or data. The color of the cable jacket shall be blue.
      - The horizontal cabling shall consist of a minimum of two (2) Category 6A UTP LP rated cables to each wireless LAN outlet. The color of the cable jacket shall be blue.
      - The length of the horizontal cables shall not exceed 90 meters (295 feet) from the telecommunications room to the work area outlet or the wireless LAN outlet.
      - Provide plenum rated cable above ceilings used as a return air plenum.
    - Workstation Outlets
      - Each Category 6A horizontal cable shall be connected to category 6A RJ45 jacks at work area outlets.
      - Each Category 6A horizontal cable shall be connected to category 6A RJ45

plugs at wireless LAN outlets.
    - The pin configuration for each RJ45 jack shall conform to the TIA/EIA T568B standard.
    - Refer to room matrix for work area outlet locations/quantities.
    - One (1) wireless LAN outlet shall be provided for each 625 square feet of floor space.
    - The typical standard density work area outlets will consist of two RJ45 interfaces. This provided connectivity for one IP telephone and one workstation.

  - Patch Cords
    - Patch cords shall be factory terminated and shall match the category of the associated patch panel, work area outlet, and horizontal cable.
  - Cable Termination Hardware
    - The Category 6A UTP horizontal cables shall be connected, in the Telecommunications Room, to Category 6A RJ45 48 port rack mounted angled patch panels. Angled patch panels containing more than 48 ports shall not be used. The pin configuration for each RJ45 jack shall conform to the TIA/EIA T568B standard.
    - The horizontal cables shall be continuous from the angled patch panels to the work area outlet jacks and wireless LAN outlet plugs.
    - The 48 port angled patch panels shall be mounted in 19-inch floor mounted equipment racks that are 84 inches tall. Wall mounted racks shall not be used except in facilities under 3,000 sq. ft.
    - Front and rear six (6) inch wide vertical cable managers shall be installed on each side of the 19-inch equipment racks on the end of the row of racks and ten (10) inch wide vertical cable managers shall be installed between each rack.
    - No more than eight (8) 48 port angled patch panels shall be installed in a single 84-inch-tall equipment rack. This allows for the lower half of the equipment rack to be used to mount Ethernet switches, UPS equipment and other network electronics. If more than eight (8) 48 port angled patch panels are required to terminate the horizontal cabling, then another equipment rack shall be installed.
- Backbone Cabling
  - Cable
  - The backbone cable from the Main Computer Room to each Telecommunications Room shall consist of a minimum of one 25-pair Category 5e UTP copper cable and 24 strands of 850 nm laser-optimized (OM4) 50/125 multimode fiber optic cable.
  - The backbone cable from the Entrance Facility Room to the Main Computer Room shall consist of a minimum of 100 pairs of Category 5e UTP copper cable and 24 strands of 850 nm laser optimized (OM4) 50/125 multimode fiber optic cabling and terminations.
  - For VA-occupied space greater than 50,000 sq. ft and for space between 25,001 sq. ft. and 50,000 sq. ft. where there are clinical services requiring dedicated medical equipment (e.g., radiology, dental, or other imaging functions), two redundant, diversely routed paths for the fiber optic backbone between the Main Computer Room and the Entrance Facility Room and the Telecommunications Rooms are required.
  - Provide plenum rated cable above ceilings used as a return air plenum.

- Provide indoor armored fiber optic backbone cable or provide unarmored fiber optic backbone cable installed in inner duct.
- Copper Backbone Cable Termination Hardware
  - The Category 5e UTP copper backbone cable shall be connected, in the Telecommunications Rooms, to 24-port rack mounted angled patch panels. (48-port angled patch panels are acceptable if more than one 25-pair backbone cable is specified.) Patch panels containing more than 48 ports shall not be used. One pairs of the backbone cable shall be terminated on each patch panel port (two pairs on port 24).
  - The RJ45 angled patch panels shall be mounted in 19-inch floor mounted equipment channel racks that are 84 inches tall. Wall mounted racks (TEs) shall not be used except in facilities under 3,000 sq. ft.
  - Front and rear six (6) inch wide vertical cable managers shall be installed on each side of the 19-inch equipment racks on the end of the row of racks and ten (10) inch wide vertical cable managers shall be installed between each rack.
  - No more than eight (8) 48 port angled patch panels shall be installed in a single 84-inch-tall equipment rack. This allows for the lower half of the equipment rack to be used to mount Ethernet switches, UPS equipment and other network electronics. If more than eight (8) 48 port angled patch panels are required to terminate the backbone cabling, then another equipment rack shall be installed.
- Fiber Optic Backbone Cable Termination Hardware
  - The multimode fiber optic backbone cables shall be connected at each end to fiber optic cable connectors in one rack position height angled high density fiber distribution panels located in the Telecommunications Room, Main Computer Room, or Entrance Facility room. The high-density patch panels shall have the capacity to terminate a minimum of 144 strands of fiber optic cabling.
  - All fiber optic backbone cable strands shall be terminated on fiber optic connectors. No fiber strands shall be left unterminated.
  - All multimode fiber optic cables shall be factory pre-terminated with Multi-Fiber Push On (MPO) connectors in the Method A (straight through) polarity configuration. The MPO connector at each end of the cable shall be connected to a fiber optic cassette to provide duplex-LC connectors at both ends that will support one or two 12-strand cable assembles or one 24-strand cable assembly for a total of 6 or 12 duplex-LC connectors. Cassettes can also be used, if directed by the VA IT department, to breakout a 24-strand backbone trunk into 3 MPO connectors of 8 fiber strands each to support 40 Gigabit Ethernet parallel transmission capabilities.
  - The fiber distribution panels shall be mounted in 19-inch floor mounted equipment racks that are 84 inches tall. Wall mounted racks and/or wall mounted fiber distribution panels shall not be used.
  - Front and rear six (6) inch wide vertical cable managers shall be installed on each side of the 19-inch equipment racks on the end of the row of racks and ten (10) inch wide vertical cable managers shall be installed between each rack.
  - No more than twelve (12) one rack position height angled high density fiber distribution panels shall be installed in a single 84-inch-tall equipment rack. This allows for the lower half of the equipment rack to be used to mount

Ethernet switches, UPS equipment and other network electronics. If more than twelve (12) one rack position height fiber distribution panels are required, then another equipment rack shall be installed.

- Installation Requirements
  - All cabling shall be installed without twists and kinks. Cables should not be looped around themselves or other objects.
  - Use cable management components and techniques to maintain clean, clear, and safe work environment. Do not mount cabling in locations that block access to other equipment inside and outside of equipment racks and cabinets.
  - Route cables with gentle loops to avoid damage due to exceeding bend radius limitations. Fiber optic cabling can be easily broken with rough handling or tight bends.
  - Cable slack should be concealed within the equipment racks and cabinets either vertically or within cable managers. Slack should not be looped. With the use of correct length cables, there should not be enough slack to require looping.
  - Patch cables should follow the side of the IT equipment rack closest to the assigned Network Interface Connection (NIC). Use correct length patch cables.
  - Label the cables, equipment cabinets and equipment racks as indicated in the VA Infrastructure Standard for Telecommunications Spaces.
- Cable Testing
  - Horizontal Cable
    - Prior to the cut-over of the equipment, test 100% of the UTP category 6A horizontal cables for performance to TIA-568-C.2, category 6A, permanent link requirements. The test instrument shall conform to the TIA-1152 Level III-e, measurement accuracy.
    - Replace and retest any cables that fail to pass the performance requirements.
    - Record the results of each test with cable identification. The test results shall be given to the VA Office of Information Technology (OIT) for each horizontal cable in electronic format.
    - The VA Project Manager shall be immediately notified if any horizontal cable fails due to link length.
  - Backbone cable
    - Copper cable
      - Prior to the cut-over of the equipment, test 100% of backbone copper cable pairs for: DC loop resistance, opens, shorts between conductors, reversed pairs, split pairs, and transposed pairs.
      - Replace and retest any cables that fail to pass the performance requirements.
      - Record the results of each test with cable identification. The test results shall be given to the VA Office of Information Technology (OIT) for each backbone cable.
    - Fiber Optic cable
      - Prior to the cut-over of equipment, test one hundred percent (100 percent) of all terminated backbone fiber strands in both directions with an Optical Power Meter and Light source to ensure the fiber strands meet or exceed the cable performance requirements of TIA/EIA-568.3-D.
      - Test instruments shall meet or exceed applicable requirements in

TIA-568.1-D. Use only test cords and adapters that are qualified by test equipment manufacturer for channel or link test configuration.

- Test multimode backbone links in both directions at both operating wavelengths of 850 nm and 1300 nm in accordance with TIA/EIA-526-14-C, Annex A, One Cord Reference Method. The tester shall be encircled flux compliant. The Channel loss shall be 2.5 dB or less for each fiber strand.
- Replace and retest any cables with fiber strand(s) that fail to pass the performance requirements.
- Record the results of each test with cable identification. The test results shall be given to the VA Office of Information Technology (OIT) for each backbone cable in electronic format.

Special Systems

- **TV Distribution System**
  - The Lessor will provide the following: Video cabling, pathways (conduit and cable tray), outlets, faceplates, amplifiers, splitters, backboards, cable terminations and cable testing.
  - The VA will provide the following: Video recorders, video signal processors, and Monitors.
  - A wired television distribution system connected to an antenna system or cable TV utility will be provided. Cabling will consist of 0.50" hardline or RG-11 trunk distribution cabling and RG6 horizontal cabling. Splitters and line amplifiers shall support 750 MHz minimum video bandwidth.
    - Splitters and amplifiers shall be located in the Main Computer Room and Telecommunications Rooms.
    - Locations: Waiting Rooms, Conference Rooms, Breakrooms, Police Operations.

- **Nurse Call System**
  - Provide tone/light nurse call system, including local audible alarm, with patient stations, toilet stations, emergency call stations, staff/duty stations, master station/annunciator, dome lights and area indicators, power supplies and additional accessories as required.
    - *Additional Nurse Call Requirements:*

    The nurse's call should be installed within reach of the patient where they would most likely need it (not necessarily location most convenient to other electrical wiring). The focus is on patient safety.

- **Access Control System**
  - The Lessor shall design, install, and maintain an Access Control System for the facility IAW the Facility Security Level.
  - Access Control equipment shall be located in Main Computer Room and Telecommunications Rooms.

- **Closed Circuit Television System (CCTV)**
  - The Lessor shall design, install, and maintain an CCTV system for the facility IAW the Facility Security Level.

- CCTV equipment shall be located in Main Computer Room and Telecommunications Rooms.

- **Intrusion Detection System (IDS)**
  - The Lessor shall design, install, and maintain an Intrusion Detection System for the facility IAW the Facility Security Level.
  - IDS equipment shall be located in Main Computer Room and Telecommunications Rooms.

- **Wireless Communications**
  - The VA space shall be served by two (2) separate wireless networks. The Guest Wi-Fi and VA Wi-Fi networks are separate
  - The Lessor shall provide and install the cabling infrastructure for the VA Wi-Fi network. The final location of the access points will be determined by the VA prior to the installation of said access point devices by the VA. The lessor shall provide one (1) Cat6a cable per 625 SF of ceiling space, one (1) cable in each corner (interior corners of exterior walls) and one (1) cable every 40 linear feet along the interior perimeter of the building. These cables shall be evenly spaced and distributed throughout the celling to provided adequate points of connections for the access points. Each cable shall be terminated in a biscuit jack. The VA may develop and provide a coverage area map noting where the Wi-Fi access points will go. In these cases, use the area map provided.
  - The Lessor shall provide Guest Wi-Fi access including installation, design, service, and operational costs. The Guest Wi-Fi system shall be designed to provide 100% coverage with established signal strength and through heat maps as identified by a wireless pre and post area coverage survey and frequency coordination study. Ensure sufficient signal strength to provide “Excellent” signal strength in the Waiting and Reception areas, and “High” signal strength throughout the rest of the Clinic Proper. Guest Wi-Fi may be unsecured and may be from common or adjacent multi-tenant space, provided the system is managed by the Lessor and is not another tenant’s signal. The guest Wi-Fi system should be separate from and with no access to VA network.

SIGNAGE AND WAYFINDING

Please follow PG 18-10 Signage Design Manual May 16, 2023 and include:

- Exterior Signage & Wayfinding To Include Main Entry Door Decal And Building Entrance (monumental) Signs
- Interior Signage & Wayfinding

JANITORIAL SERVICE:

- The Lessor shall maintain the Premises and all areas of the Property to which the Government has routine access in a clean condition and shall provide supplies and equipment for the term of the Lease.
- Response time of 1 hour during normal business hours for emergency janitorial services. Cleaning to be performed as per OSHA requirement for Bloodborne Pathogens and/or CDC requirement of Environmental Service.
- Lessor to provide unscheduled terminal cleaning when deemed necessary from infection control.
- Terminal Cleaning: The purpose of this cleaning and disinfection process is to remove bacterial contamination from environmental surfaces and equipment surfaces where patients receive care to prevent the transmission of the microorganism from patient to patient, from patients to healthcare

workers, and from patients to visitors.

- Terminal Cleaning services personnel shall receive proper training and education on patient room cleaning and disinfecting protocols, and they must use all precautions (such as masks, gloves, and gowns) identified in the Personal Protective Assessment when cleaning in rooms or units where surfaces may be contaminated with infectious microorganisms. All training will be coordinated with the VA infection control staff and will be completed within 30 calendar days of contract award.
- Terminal Cleaning services personnel shall use Environmental Protection Agency (EPA)-approved, hospital-grade cleaning and disinfectant products and all horizontal surfaces and other surfaces in the room that may have become contaminated must be cleaned and disinfected. Contractor is to provide all terminal cleaning supplies.
- Contractor shall clean all high-touch areas in the room including cabinets, workstations and inner drawer, phone and cradle, armchairs, door and cabinet handles, light switches, etc.
- Privacy curtains should be removed by the Contractor, placed in a plastic bag in the room and double bagged into a laundry bag with the assistance of another member of the terminal cleaning services staff standing at the door outside the room. The person outside the door should wear gloves. After completing the task this person should remove gloves, wash hands with an antimicrobial soap and water or apply an alcohol rub to their hands. The terminal cleaning shall be completed following the Standard Operating Procedure developed by the user.
- If visibly soiled, Contractor shall clean window curtains, ceiling, and walls.
- Following the terminal cleaning of a patient room, gloves should be removed to avoid touching the outside of the gloves. Hands should be washed with an antimicrobial soap and water, or an alcohol rub applied to the hands prior to donning a new set of gloves.
- The Contractor shall use Green cleaning products and processes including, but not limited to products containing recycled content, environmentally preferable products and equipment, vacuum cleaners with High Efficiency Particulate Air (HEPA) filtration, bio-based products, and products and equipment that minimize the use of energy, water, and other resources. All cleaning cloths and mopping programs used at direct patient care areas, at a minimum; shall be of microfiber technology.
- An inventory list of products to be used under this contract, along with the respective SDS shall be provided to the COR for evaluation and approval upon contract award. The list shall include manufacturer's name and brand name. The list shall be updated, throughout the term of the contract. Contractor shall maintain the SDS in a location accessible to all employees and shall advise the COR of such location.
- No material or supplies shall be used in the performance of this contract or placed or stored on government property until the applicable SDS have been furnished to the Contracting Officer/COR and has been approved. Costs for correcting damage caused by misused materials will be borne by the Contractor.
- Standard Services: The contractor shall clean all spaces in accordance with customary hospital-grade, aseptic techniques standards and VA approved contractor procedures manual. A hospital-grade germicide shall be used in all areas. Standard precautions shall be always practiced. There may some cases where a room requires a "terminal Cleaning" in the case of a sick patient or otherwise. Upon request from VA staff, the contractor shall perform a terminal clean to affected areas.
- Performance Standards: Through innovation, technology, or other means the Contractor shall perform the work in this contract to meet the quality and performance standards in this section. Evaluation of the Contractor's work shall be based on the standards in this Section and conducted in accordance with the Government's Quality Assurance Surveillance Plan (QASP)

- Davis Bacon Act Wages must be utilized for the build out of the lease space.
- As part of JCAHO inspection standards, Sites must show that at least 95% of the program's items work properly at any given time. Exit Lights and Lights in the means of egress must be in working condition at all times. Inspect smoke barrier penetrations and fire rated corridor wall penetrations quarterly for penetrations and repair with appropriate fire-rated materials. Onsite lessor rep should be inspecting the facility monthly to assess fire rated door latches, panic hardware, fire extinguishers, eye wash stations, ensure corridors are clear, all common areas assessed for damage, building systems are fully operational, etc. During repairs or maintenance, a process is in place to manage air quality, infection control, utilities, noise, odor, dust, vibration, and other hazards; high-risk and infection control preventive maintenance completion rates at 100 percent; and non-high-risk completion rate at 90 percent.

EMERGENCY RESPONSE:

- Past emergencies, outages and power disruptions have shown the Government that equipment such as portable A/C chillers, industrial fans, water pumps, complete structure dry-out gear, high volume air movers, high volume dehumidifiers, mold or sewage remediation and high-volume floor/carpet dryers may be required to restore services to ensure the continuation of normal facility services. It is the Lessors' responsibility to have immediate access to any equipment that may be needed in response to an emergency or utility outage.
- The lessor shall be responsible for ensuring that the HVAC (Heating, ventilation, and air-conditioning) system are always fully operational and capable of providing a temperature range per VA Design Manual. Temperatures must be maintained throughout the leased premises and service areas, regardless of outside temperatures and the opening and closing of exit doors, during hours of operation specified in the lease. In the event of a utility system failure it is the lessor's responsibility to respond to the following:
- The Lessor shall have a building superintendent or locally designated representative available to promptly address and correct any deficiencies that may impact use of the building.
- When the Lessor is notified that his/her action is required to correct any deficiencies, the lessor will respond within one hour via telephone or e-mail with a plan of action and an expeditious time frame to resolve the deficiency(ies)." The Government reserves the right of approval of the plan of action and an expeditious timeframe to resolve the deficiency(ies).
- Four-hour response time on any utility or building system problems, to include but not limited to a/c, plumbing, electric, roof leaks, etcetera. Repair personnel shall be on site to effect repair within four hours or appropriate time frame determined by the government. If the deficiency(ies) is not corrected within the government approved timeframe for plan of action, the Government will exercise their right under GENERAL CLAUSES No 15, 48 CFR 552.270-10, Failure in Performance, and may have the deficiencies corrected at the Lessors expense. The dollar amounts for any equipment or services necessary to respond to or correct deficiencies will be deducted from the Lessor's rent.
- Non-urgent calls, as determined by the COR, will be addressed within 24 hours of the facility manager being notified. If no evidence of action being taken 24 hours after the problem has been reported; the Contracting Officer will have the option to Contract the Services of an outside Contractor, order the repairs directly and deduct the total cost of the repairs from the base monthly rental rate.
- The lessor will provide a 24-hour emergency point of contact phone number that will be responsible for all emergencies during normal and after work hours including weekends and holidays.
- The lessor shall submit a building emergency plan annually that clearly defines the actions the lessor will take in the event of emergencies. It will clearly define the details of what actions will be taken by the lessor in the event of:
  - Loss of water
  - Loss of Electricity
  - Loss of Heating and cooling
  - Structural damage to the building
  - Interior water intrusion containment, response and recovery
  - Smoke and water damage remediation
  - Flooding from a ruptured supply or flow pipeline within the building
  - Disruption of heating, ventilation, and air conditioning services (HVAC) services
  - Release of airborne particulates such as dust, mold asbestos and fiberglass that may be hazardous to building occupants.

QUALITY ASSURANCE

- **Quality Control:** The Contractor shall develop and maintain a quality program to ensure all services are performed in accordance with commonly accepted commercial practices. The Contractor shall develop and implement procedures to identify, prevent, and ensure non- recurrence of defective services by conducting frequent inspections with the COR. As a minimum the contractor shall develop quality control procedures addressing the areas identified in Basic Cleaning Services, Basic Restrooms and Shower Cleaning Services, Exterior Cleaning, Periodic Cleaning Services, and Response to Emergency Repairs and Special Event Cleaning Services as well as Detailed Response to Unforeseen Events, Times and Schedules, to the CO for acceptance not later than the pre-performance conference. The CO will notify the Contractor of acceptance or required modifications to the plan before the contract start date. The Contractor shall make appropriate modifications and obtain acceptance of the plan by the Contracting Officer before the contract start date.

- **The plan shall include:**

  A description of the inspection system to cover all services specified in the statement of work. Description shall include specifics as to the areas to be inspected on both scheduled and unscheduled basis, frequency of inspections, and the title and organizational placement of the inspectors. Additionally, control procedures for any government provided keys or lock combinations should be included.

  - A description of the methods to be used for identifying and preventing defects in the quality of service performed.
  - A description of the records is to be kept to document inspections and corrective/preventive actions taken. Copy to be provided to the COR
  - The records of inspections shall be kept and made available to the Government throughout the contract performance period and for the period after contract completion until final settlement of any claims under this contract.
  - Contractor Quality Control Program: Contractor shall have a quality control program to assure all requirements of the contract are provided as specified. The program shall be continuously improved and is therefore documented in loose-leaf manual format. The program shall include, but not be limited to the following:
  - Written work instructions/procedures, processes, and product descriptions to implement contractual obligations. The preparation and maintenance of and compliance with, these instructions shall be audited as a function of the Contractor's Quality Control Program to assure compliance with or timely changes to the instructions. The COR shall be on document distribution for all formalized changes to the Contractor's Quality Control Program. The COR will request corrective action to improve the quality of patient care or cure damage to the facility. Written work instructions will be complete and reliable. Reliable records are objective evidence of the past quality of service.
  - A method of early detection and correction of assignable conditions adverse to the quality of service, to include analysis or corrective action records (including customer complaints) to determine causes of defects. This method will include providing timely written explanation/documentation of the correction of the defectiveness and correction of cause in response to Government’s COR reactive action request to include bacteriological monitoring when necessary.

QUALITY CONTROL MONITORING

- The Government appointed COR will monitor the Contractor’s performance to assure that the performance thresholds and standards of performance are met. In accordance with FAR 52.212-4
- “Inspection/Acceptance” the Government reserves the right to inspect or test any supplies or services that have been tendered for acceptance. The Government may require repair or replacement of nonconforming supplies or performance of nonconforming services at no increase in contract price.
- The Contractor service requirements are summarized into performance objectives that relate directly to mission essential items. The performance standards describe the minimum acceptable levels of the service required for each task. These thresholds are critical to mission success.

- The Government's COR will evaluate the services required by this contract to ensure compliance and quality
- The contractor shall perform all work required by this contract in a satisfactory manner in accordance with the SOW. The COR will not consider the task complete until all deficiencies have been corrected.
- The Government's COR will inspect all work tasks required by the task order sheets to ensure contract performance on a monthly basis utilizing the attached work inspection sheet.
- The Government's COR will receive complaints from facility personnel and pass them on to the Contractor's quality inspector for correction.
- The inspection period is weekly. Inspection period will be from the first of the month through the last day of the month. The COR should receive no more than five (5) complaints. The COR will record results of the inspection, noting the date and time of inspection. If an inspection indicates unacceptable performance, the COR will notify the supervisor or quality inspector. The Contractor shall be given four(4) hours after notification during shift hours to correct the unacceptable performance. Report period is weekly; however, complaints are by task.

APPENDIX A

1. **BUILDING CODES AND STANDARDS.**

Construction Documents shall incorporate Civil (site), Architectural, Mechanical, Electrical, & Plumbing (MEP), Structural, components necessary to provide for a complete structure and functioning building, following BUILDING CODES AND STANDARDS:

PG-18-3 Design and Construction Procedures	https://www.cfm.va.gov/til/cPro/cPro.pdf
PG-18-5 Equipment Guide List	equip263.xlsx
PG-18-9 Space Planning Criteria	https://www.cfm.va.gov/til/space/spChapter263.pdf
*Fire Protection Design Manual	dmFire FIRE PROTECTION DESIG
*Signage Design Manual	https://www.cfm.va.gov/til/dManual/dmSignage.pdf
*Interior Design Manual	https://www.cfm.va.gov/til/dManual/dmIDhonh.pdf
*HVAC Design Manual	https://www.cfm.va.gov/til/dManual/dmHVAC.pdf
*Electrical Design Manual	https://www.cfm.va.gov/til/dManual/dmElec.pdf
*Lighting Design Manual	https://www.cfm.va.gov/til/dManual/dmLighting.pdf
*Plumbing	https://www.cfm.va.gov/til/dManual/dmPlbg.pdf
Commissioning	Whole Building Commissioning Proc

LESSOR: ________ GOVERNMENT: ________

PG-18-14 Room Finishes, Door, and Hardware Schedule	https://www.cfm.va.gov/til/room/RoomFinishes.pdf
Barrier Free Design Standard	dsBarrFree (VA BARRIER FREE DESIG
Architectural Barriers Act Accessibility Standard (ABAAS) for Federal Facilities	https://www.access-board.gov/files/aba/ABAstandards.pdf
Bariatric Safe Patient Handling and Mobility	BariatricSafePatientandHandlingGUIDE
Safe Patient Handling and Mobility Design Criteria	dcSPHM SAFE PATIENT HANDLING
003C2B-DA-149 Design For Patient Privacy And Women Veterans' Health	dAlert149_women and privacy.pdf
UPS General Chapter <797> Hazardous Drugs – Handling in Healthcare Settings from USP 40 – NF 35, Second Supplement (2017)	
USP General Chapter 795 Pharmaceutical Compounding – Nonsterile Preparations	
H 18-8 Seismic Design Handbook	18 8 seismic design requirement, VA har
VA Enterprise Facility IT Support Infrastructure Standard	
The Joint Commission (TJC) accreditation standards apply to the facility under the affiliated VA medical center license. Building construction and on-going maintenance procedures shall meet TJC standards. Lessor shall provide and submit all documentation that is required for TJC requirements.	
NFPA 99: Health Care Facilities Code	
NFPA 101: Life Safety Code	

LESSOR: ________ GOVERNMENT: ________

The Facility Guidelines Institute Guidelines for Design and Construction of Health Care Facilities, as reference using more stringent elements found in VA Design Guides and Manuals as listed above.	
Interagency Security Committee (ISC) Risk Management Process	
Facility Security Level (FSL) shall be as outlined in GSA Form L100 Global Lease.	

2. REFERENCES:

- National Electric Code (NEC), part 800 article 250
- Building Industry Consulting Service International (BICSI) standards.
- Electronic Industries Association / Telecommunications Industry Assoc. EIA / TIA 569 Standard for telecommunications pathways and spaces requirements
- State/City/Local building ordinances and codes as applies to, include Building Code
- Telecommunications Industry Assoc. / Electronic Industries Association TIA / EIA 606 (Standard for records, labeling and space & pathway administration)
- Installation of outside plant, inside riser, and station cabling shall conform to meet the requirements of ICEA Publications S-80-576-1988 (Ref.B1.6) as to size and installation practice.
- The installation of the cable shall conform to appropriate OEM, ANSI/EIA/TIA recommendations, Federal Communications Commission (FCC) part 68
- Underwriters Laboratories (UL)

SAMPLE FLOOR PLAN